CANCELLATION FOR QUASI-COMPACT MORPHISMS OVER A QUASI-SEPARATED BASE

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 4590-4601, following the statement at 4583-4588.

Allowed standard prerequisites (editorial).

- Sections of qcqs morphisms are immersions. lemma-section-immersion (same chapter, cited at ~4593). (1, f): X -> X x_Z Y is quasi-compact.
- Base change preserves quasi-compactness. lemma-quasi-compact-preserved-base-change (same chapter, cited at ~4598). X x_Z Y -> Y is quasi-compact.
- Compositions of quasi-compact morphisms. lemma-composition-quasi-compact (same chapter, cited at ~4600). Concluding composition step.
- Separation axioms stable under base change. lemma-separated-permanence (card L26393, cited at ~4595). X x_Z Y -> X is quasi-separated as a base change of g.

Source reference. Original lemma 01K6: sources/stacks-schemes/pinned-original-source.tex, source lines 3853–3853. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 01K5: sources/stacks-schemes/pinned-original-source.tex, source lines 3843–3843. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 01KT: sources/stacks-schemes/pinned-original-source.tex, source lines 4492–4492. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 01KU: sources/stacks-schemes/pinned-original-source.tex, source lines 4506–4506. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $f : X \to Y$ and $g : Y \to Z$ be morphisms of schemes. If $g \circ f$ is quasi-compact and g is quasi-separated then f is quasi-compact.*

Date: stacks-project:schemes.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Proof. This is true because f equals the composition $(1, f) : X \to X \times_Z Y \to Y$. The first map is quasi-compact by Lemma 01KT because it is a section of the quasi-separated morphism $X \times_Z Y \to X$ (a base change of g, see Lemma 01KU). The second map is quasi-compact as it is the base change of $g \circ f$, see Lemma 01K5. And compositions of quasi-compact morphisms are quasi-compact, see Lemma 01K6. $\square$